HakaseAI | BIODIGITAL TWIN

THE PRE-CLINICAL BIODIGITAL TWIN · PRODUCT DOCUMENTATION

Predict your IND-ready package before you fund the studies.

Hakase BioDigital Twin walks a candidate molecule across four pre-clinical layers — from in-vitro screen to a sealed, IND-ready handoff — entirely in silico. Think of it as an extra pair of expert hands for your team: **runway extension, not lab replacement**, helping you point your bench toward the molecules that truly deserve it.

L1 · IN VITRO TWIN | L2 · CONFIRMATION | L3 · ANIMAL COHORT | L4 · FIRST-IN-HUMAN READINESS | PRIVACY-FIRST

Hakase BioDigital Twin · Candidate molecule → sealed, IND-ready handoff.
Zero-lab cycle time. Regulator-credible evidence. Privacy-first by design.

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For client & partner distribution

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01 What the BioDigital Twin is

A working model of the pre-clinical journey, run before the lab

Hakase BioDigital Twin is a pre-clinical simulation platform. A scientist enters a candidate molecule and its target, and the platform walks that molecule through four cascading layers — from in-vitro screen, through confirmation and cross-species behaviour, to a defensible first-in-human starting dose — and seals the result into an IND-ready handoff. The entire cascade runs in silico, with no wet lab required to produce the prediction.

The purpose is simple: help you answer the questions that decide a molecule's fate *before* you spend months and budget finding out at the bench. The Twin does not sign off your candidate or run your assays — it surfaces which molecules deserve your bench time, and which were never going to make it, on day one instead of month fourteen.

We built it the way a scientist would want a tool built: it shows its work, explains its reasoning, and always leaves the final call with you. You stay in command of the science — the Twin just makes sure you are looking in the right place first.

90%	$2–5M	12–18mo	0 bytes
of drugs fail in the clinic	spent per IND filing	to pre-clinical sign-off	of structural IP transmitted

What it is — and what it is not

The BioDigital Twin is…

- \+ A **decision layer** that ranks candidates before the bench is engaged.
- \+ A way to **extend a biotech's runway** by spending lab budget only on molecules that pass.
- \+ An **early-warning system** for toxicity, absorption and dosing problems.
- \+ A producer of a **sealed, regulator-ingestible** pre-clinical package.

The BioDigital Twin is not…

- – A **replacement** for your scientists, your lab, or your CROs.
- – An **autonomous decision-maker** — a human owns every override.
- – A **substitute** for the real experiments that confirm a candidate.
- – A **data-harvesting service** — your molecule never leaves your control.

02 The pre-clinical problem

Three questions decide whether a molecule ever reaches a human

Before a small biotech can dose a single person, it has to answer three pre-clinical questions — and answer them defensibly enough that an IND filing holds up to a regulator.

1 **Is this molecule worth confirming?**

Is it absorbable, drug-like, and free of obvious organ-toxicity liabilities — based on real chemistry, not back-of-the-envelope estimates?

2 **Does the verdict survive deeper confirmation?**

Does structure-based confirmation agree with virtual-tissue projection of the in-vitro screen — and does the candidate behave safely across pre-clinical species?

3 **Are we IND-ready, and what is a defensible first-in-human dose?**

Have we covered cross-species pharmacokinetics, safe-dose thresholds and the IND-enabling checklist — and can we justify a conservative starting dose?

> THE HARD TRUTH
>
> Answering these three questions the traditional way costs **$2M–$5M** and takes **12–18 months** of lab experiments, animal studies and consulting. And roughly **90% of drugs that enter clinical trials still fail** — most for reasons that could have been seen earlier: poor absorption, unexpected toxicity, weak target engagement, or a starting dose that was wrong from day one. By the time the problem surfaces at the bench, the money is already gone.

The wet lab is precious — and finite

The bottleneck is rarely the experiment itself; it is choosing *which* experiments to run. A small team must triage many plausible molecules against limited budget and time, and good candidates are routinely starved while resources are spent confirming molecules that a deeper first look would have set aside. Every wasted cycle is runway a biotech cannot get back.

03 What we built — a flight simulator for molecules

The model doesn't replace the flight; it tells you whether to make it

Before a pilot flies a real plane, they spend hundreds of hours in a flight simulator. The simulator doesn't replace real flying — but it tells you whether *this* pilot, in *this* plane, on *this* route, is likely to succeed. And it does so without risking a single life or a dollar of jet fuel.

Hakase BioDigital Twin does the same for the pre-clinical journey. A scientist enters a molecule and a target, and four cascading layers walk it from in-vitro screen to a sealed, IND-ready handoff — every output tagged with how it was produced and how confident the platform is, every human override logged. The bench remains the place where reality is confirmed; the Twin simply makes sure the bench is pointed at the right molecule.

- **Decisions in seconds**

  A safety or absorption flag that would take months to surface at the bench appears in seconds, on day one — while it is still cheap to change course.

- **Confidence on every output**

  Each result carries a confidence level and a source-quality tier, so a scientist knows when to trust a verdict and when to send it to the bench.

- **The human stays in command**

  Nothing is decided silently. Every flag is traceable and every override is owned by a scientist or clinician — the Twin advises, people decide.

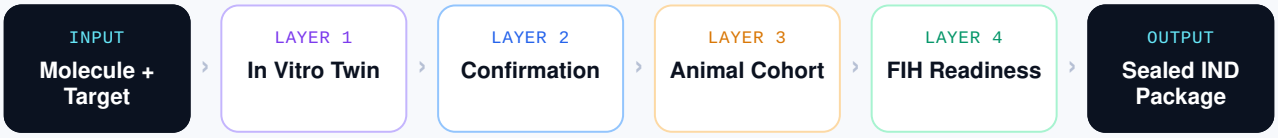
"We don't replace scientists or labs. We make sure every dollar a small biotech spends on real experiments is spent on the right molecule."

— HAKASE

04 The four-layer cascade

One candidate, four gates, a single sealed package

The platform is organised as a cascade. A candidate is evaluated layer by layer, and only advances where the evidence supports it. Weak molecules are filtered early and cheaply; the few that reach the top arrive with a complete, defensible evidence trail behind them.



EACH LAYER GATES THE NEXT · EVERY OUTPUT TAGGED WITH CONFIDENCE & PROVENANCE

Because every layer shares one representation of the candidate, the result is a single coherent profile — from molecular plausibility to a defensible human starting dose — that a scientist can read end to end and a regulator can ingest. **The throughline is simple: test a hundred molecules before lunch in silico, then spend real lab money only on the handful that clear all four gates.**

05 Inside the four layers

What each layer asks, does, and hands forward

L1 In Vitro Twin

5–30 SEC

Is this molecule worth confirming?

A fast first-pass screen on real chemistry: drug-likeness and molecular properties, a full ADMET profile with confidence intervals, target-binding affinity, and an organ-toxicity check — returning a clear gate verdict with a quality tier on every output.

- GO / WATCH / NO-GO
- ADMET + confidence
- Binding affinity
- Organ-tox flags

L2 Confirmation Layer

1–10 MIN

Does the verdict survive deeper confirmation?

Two parallel computational tracks pressure-test the L1 verdict — a structure-based track (binding poses, selectivity, cardiac-safety, optional physics-grounded refinement) and a virtual-tissue ex-vivo read with no physical tissue — sealed into one tamper-evident block.

- Structure-based confirmation
- Selectivity & cardiac-safety
- Virtual-tissue read
- Sealed block

L3

Animal Cohort

10–60 SEC

How does it behave across species?

A physiology-based model projects behaviour across standard pre-clinical species, with formulation modelling and regulator-aligned scaling. It returns species-by-species safe-dose thresholds, safety margins, and a human-equivalent dose — surfacing any mismatch as a flag, not a silent fix.

- Multi-species PK
- Formulation modelling
- NOAEL + safety margin
- Human-equivalent dose

L4

First-In-Human Readiness

INSTANT

Are we IND-ready?

Derives a defensible, conservative first-in-human starting dose aligned to FDA/EMA guidance, runs the IND-enabling gap analysis — toxicology, manufacturing readiness, investigator's brochure, pre-IND — proposes a Phase 1 plan, and seals everything upstream into a regulator-ingestible package.

- Conservative FIH dose
- IND-enabling gap analysis
- Phase 1 plan
- Sealed package

06 HAIOps — the governance layer

What keeps every prediction honest, traceable, and human-owned

A simulation is only useful if you can trust it — and trust, in drug development, comes from governance. HAIOps is the layer that runs across all four cascade layers and holds the whole platform to account. It is the reason a prediction from the BioDigital Twin is something you can put in front of a sponsor or a regulator, rather than a score you have to take on faith.

Every output the platform produces is tagged with **how it was produced and how confident the platform is** — a source-quality tier that tells a scientist when to rely on a number and when to send it to the bench. Every flag is traceable back to the evidence behind it, and every human override is recorded. Nothing is decided silently, and nothing advances unless the gate is satisfied.

Provenance on every output

Each result carries a source-quality tier and confidence — so a scientist always knows whether a number is solid or worth confirming at the bench.

Predict-then-confirm, enforced

A candidate only advances where the gate is satisfied. Nothing is waived quietly, and re-running a sealed step requires an explicit, recorded unseal.

Every override is owned

Flags trace to their evidence and every human decision is logged. The platform advises; a scientist or clinician always makes the call.

One signed audit trail

It all seals into a tamper-evident, exportable provenance trail behind the IND-ready package — the same record a sponsor or regulator can review.

WHY GOVERNANCE MATTERS

HAIOps is the difference between a result you have to believe and a verdict you can defend. It is what lets a small team move at in-silico speed without giving up the rigour a regulator expects.

07 Why it matters — every dollar on the right molecule

For a small team or lab, every dollar and every week counts

A small biotech typically raises **$5M–$15M** in seed or Series A funding. That money has to cover everything — the team, the lab work, regulatory preparation, and one real shot at a clinical trial. Where that capital goes decides whether the company gets a second shot at all.

WITHOUT THE BIODIGITAL TWIN

The company spends $2M on lab experiments, only to discover at month fourteen that the molecule carries a fatal cardiac-safety issue. The program is over. The money — and the runway — are gone.

WITH THE BIODIGITAL TWIN

That same safety flag surfaces in seconds, on day one. The team screens a hundred molecules before lunch and commits real lab budget only to the candidates that clear all four gates.

The Twin doesn't make the wet lab unnecessary — it makes the wet lab *count*. By moving the expensive discovery of a fatal flaw from month fourteen to day one, a fixed budget produces far more validated leads, and a fixed runway lasts far longer.

08 Outcomes & value

What you actually get

Cut decision time

From candidate to IND-readiness in days, not quarters — compressing months of pre-clinical work into a single defensible package, without expanding the wet-lab footprint.

Defensible at every gate

Conservative first-in-human dosing, regulator-aligned cross-species scaling, and a complete provenance trail — built to survive sponsor and regulator scrutiny.

Privacy by construction

Molecular structures stay on your machine. The platform only ever processes one-way derived signals — there is no path back to your candidate.

30–50%	100s	Day 1
Faster pre-clinical cycle	**Molecules screened**	**Fatal flaws surfaced**
Decision time compressed	Before committing lab budget	Not month 14

THE BOTTOM LINE

The BioDigital Twin turns pre-clinical from a slow, capital-burning gamble into a fast, evidence-led triage — so scarce runway is spent advancing the molecules most likely to reach a human.

09 Privacy & intellectual property

Your molecule is your most valuable asset — it stays yours

A drug candidate's structure is the single most valuable piece of IP a biotech holds. Hakase BioDigital Twin is built so that using the platform never means handing that structure over.

Structural IP stays client-side

The first-pass screen runs locally on your own machine. Your molecule's structure is never uploaded.

One-way derived signals only

Deeper layers operate on derived signals — never the structure itself. There is no path back to your candidate from anything the platform receives.

Your bench data stays yours

In-house experimental data you choose to add improves your results only. It never trains a shared, global model.

Traceable & tamper-evident

Every prediction and override is logged into a sealed, exportable provenance trail you can review and defend.

IN PLAIN TERMS

It's like having a multi-million-dollar lab inside your laptop, with the confidentiality of a sealed notebook — zero bytes of structural IP ever leave your environment.

10 How you engage — Drug-Acceleration-as-a-Service

Outcomes, not seat licenses

The BioDigital Twin is delivered as **Drug-Acceleration-as-a-Service (DAaaS)** — an engagement model built around outcomes rather than software seats. You pay for the pre-clinical deliverables you actually use: per molecule, per stage, per sealed package, aligned to the milestones your program already tracks. There is no shelfware to absorb and no per-seat licensing to negotiate.

That model scales in both directions. A two-person team can run a handful of candidates this quarter; a larger group can run thousands on the same engine — and in every case the cost follows the work, not the size of the team. It turns pre-clinical evaluation from a fixed capital commitment into a variable cost you can switch on exactly when a decision needs it.

Pay per deliverable

Priced per molecule, per stage, per sealed package — mapped to the milestones you already track. No seats, no shelfware.

Scales with the work

Run a few candidates or thousands on the same engine; cost follows the work, not the headcount.

Private by default

Every engagement is tenant-isolated, your structural IP stays client-side, and the engine never trains on your data.

Start small, expand later

Begin with a single program and grow across your pipeline as the evidence earns your trust.

11 Who it's for & where it fits

From lean biotechs to academic labs and research institutes

The BioDigital Twin slots into the discovery pipeline you already run. It accepts a candidate from any source — generative chemistry, virtual screening, or in-house medicinal chemistry — and adds a decision layer at exactly the points where scarce budget and bench time are allocated.

It is a particularly natural fit for **academic and translational research**. Grant-funded labs operate under exactly the constraints the platform was built for — tight budgets, limited bench time, and the need to show defensible pre-clinical evidence to funders, partners and prospective spin-out investors. The same zero-lab cascade that extends a biotech's runway lets a university group screen far more ideas per grant, and gives students a clear, end-to-end window into how a molecule travels toward the clinic.

Small & emerging biotech

Stretch a seed or Series A round by de-risking candidates before committing to costly studies.

Academic & university labs

Screen far more ideas per grant, and give students an end-to-end view of the pre-clinical journey.

Research institutes & cores

Offer many investigators a shared, governed pre-clinical screen without standing up new lab capacity.

Discovery & med-chem teams

Triage many molecules quickly and focus assays on the candidates most likely to confirm.

Translational & pre-clinical leads

Carry one coherent profile from screen to human-equivalent dose into IND-enabling decisions.

Tech-transfer & spin-outs

Judge the IND-readiness of lab candidates to decide what's worth advancing, licensing, or spinning out.

Founders & R&D leadership

Allocate finite runway toward the molecules with the best odds of reaching a human.

Regulatory & CMC

Assemble an early, defensible IND-enabling picture with a complete provenance trail.

CRO partners

Receive sharper, pre-prioritised experimental plans instead of an undifferentiated queue.

12 Get started

We'd love to run your molecule through the cascade

The best way to get a feel for the BioDigital Twin is to watch it walk one of your own candidates across all four layers — the verdict, the confidence on every output, and the sealed package at the end — so you can weigh it against your own bench, not a slide. We'll set it up around a target or molecule you actually care about. No pressure and no obligation: just a hands-on session and an honest look at the science, together.

LET'S TALK — WE'RE EASY TO REACH

Bring us a molecule you believe in.

Whether you're triaging candidates, preparing for IND, stretching a grant, or simply curious how your lead holds up, we'll walk it through with you, step by step. Tell us what you're working on — we read every message, and we love talking shop.

EMAIL US ANYTIME
info@hakase.ai

REQUEST A DEMO
A guided run on your molecule

PARTNER WITH US
Co-develop on a live program

Questions, ideas, or just want to see it in action? Write to us at **info@hakase.ai** — we'd be glad to hear from you. Figures in this document (cycle-time, cost and screening throughput) are indicative of typical platform impact and not a guarantee of results for any particular program; scope and outcomes are defined per engagement. For client and partner distribution.